Fiscal Year:	FY 2021	**Task Last Updated:**	FY 11/21/2020
PI Name:	Steen, Paul H Ph.D.		
Project Title:	Capillary Flows and Draining in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	14853-5201	**Congressional District:**	23
Comments:	PI is deceased as of September 2020		
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2017 Physical Sciences NNH17ZTT001N-17PSI-D: Use of the NASA Physical Sciences Informatics System – Appendix D
Start Date:	01/31/2019	**End Date:**	01/30/2021
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Hatch, Tyler	**Contact Phone:**	216.433.5073
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	NOTE (Ed., November 2020): Principal Investigator Dr. Paul Steen passed away in September 2020; Dr. Olivier Desjardins will now oversee Joshua McCraney's research on this project. NOTE (Ed., Mar 2019): This work is result of proposal by graduate student Joshua McCraney, Ph.D. candidate at Cornell University, who will be performing the work with advisor PI Dr. Paul Steen.		
COI Name (Institution):	McCraney, Joshua M.S. (Graduate Student PI: Cornell University)		
Grant/Contract No.:	80NSSC19K0406		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Capillary Flow Experiments (CFE) for Interior Corner Flows (ICF) were conducted aboard the International Space Station (ISS) and rely solely on video for high-quality quantitative data. We propose to digitally measure the still video images of the time-dependent interface height. Since container geometry is known, the meniscus heights can be used to determine volumetric drain rates. This data would then serve as both a benchmark for current theory on drain rates and a guide for extending capillary flow models to a diverse set of geometries and flow conditions. We also propose to utilize and extend a numerical tool to simulate 3D transient corner flows with and without topological change. To pursue any discrepancy between data and numerics, supportive ground-based experiments at micro-scales and low-gravity experiments conducted in the Dryden Drop Tower at Portland State University are proposed. Understanding the dynamics of capillary flows and responses of drain flows to various geometries offers design advantages for spacecraft waste-management systems, fuel control, hydration containment, and cryogenic flows. Capillary flows and drain problems are relevant to a broad range of two-phase flow technologies in low gravity, including 3D vane networks for positioning of fuels, an important problem concerning propellant liquid management aboard spacecraft.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The capillary drain flows see immediate terrestrial application for biofluids during lab-on-chip processing. For low-gravity environments, capillary-dominated fluid behavior is common over large length scales in the near-weightless environment aboard spacecraft. In part due to both the expense and limited opportunities for designers to develop significant intuition of general multiphase phenomena in such environments, developing robust fluid systems for spacecraft fuels management, thermal control, and water processing for life support remains challenging. For the specific example of liquid fuels aboard orbiting spacecraft, capillary draining can serve as a limit to the life of the spacecraft if and when the residual fuel in the tank is or becomes inaccessible. It is essential to establish the maximum capillary flow rate at which a container in a micro-gravity environment can be drained to increase satellite predictability and lifespan (when fuel is empty, the satellite is essentially expired).
Task Progress:	Joshua McCraney is working on two over-all large projects: 1) low-gravity fluid drain problems emphasizing numerical and experimental methods and 2) contact line phenomena emphasizing theoretical techniques. The first suite of projects, McCraney virtually attended the 15th OpenFOAM Workshop in June 2020. He presented on low-gravity fluid draining in wedge geometries. This work compared theoretical, experimental, and numerical work. McCraney developed a numerical tool in OpenFOAM (a computational fluid dynamics CFD flow solver) and programmed the simulations to emulate flight experiments from the Physical Sciences Informatics (PSI) video archive, specifically the Interior Corner Flow (ICF) tests. With this work, McCraney published a conference proceeding to the workshop. McCraney then published an extension of this work to a journal. Prior to this work, the particular experiments studied had never been analyzed or published. McCraney has further mined the ICF database and identified several flows of interest. McCraney then digitized the time-dependent interface profiles as still images, which are collected from the ICF data with which comparisons to analytical and numerical predictions are immediate. The meniscus is tracked and analyzed by a new in-house tracking algorithm recently designed and implemented by McCraney. He has also identified several single drains through a myriad of geometries and passive bubble migration flows. Some of these flows have been regressed from video to still images, readily analyzed in the same technique already published by McCraney. McCraney has fully analyzed a few of these flows, and is currently comparing them against simulations and developed theory. The second suite of projects, McCraney has implemented a new technique to sweep fundamental frequencies for low-gravity fluids held in a support geometry. This work will be discussed virtually at the Annual Physical Society (APS) Division of Fluid Dynamics (DFD) conference November 22-24, 2020. McCraney has a detailed manuscript of this work, and anticipates submitting for publication to the journal Society for Industrial and Applied Mathematics (SIAM) by the end of the year. Within this second project suite, McCraney is formulating the mathematical constructs to analyze the fundamental frequencies of liquid contained in a right circular cylinder. The bulk of the spectral techniques have been coded, and results will be compared to already conducted terrestrial experiments made by external universities.
Bibliography Type:	Description: (Last Updated: 09/04/2025)
Articles in Peer-reviewed Journals	McCraney J, Weislogel M, Steen P. "OpenFOAM simulations of late stage container draining in microgravity." MDPI Fluids. 2020 Dec;5(4):207. https://doi.org/10.3390/fluids5040207 , Dec-2020
Papers from Meeting Proceedings	McCraney J, Steen P. "Capillary draining in zero-g: the double drain problem." Conference paper peer-reviewed and published at the 15th OpenFOAM Workshop, Virtual, June 22-26, 2020. 15th OpenFOAM Workshop, Virtual, June 22-26, 2020. , Jun-2020